CS 550/ECE 552 Project Status Report: Replication of the Hawkeye and Mockingjay Cache Replacement Policies

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Abstract—This report describes our progress in replicating two cache replacement policies that approximate Belady's optimal algorithm, Hawkeye and Mockingjay, in gem5. Hawkeye is implemented in gem5, and its OPTgen matches a reference model on all 9,218 decisions of a test log. We evaluated it on six SPEC CPU2017 benchmarks and three last-level cache (LLC) sizes. It reduces LLC misses by 6.3% on mcf, the benchmark with the most LLC misses, but increases them on omnetpp and gcc. We traced most of this increase to instruction lines that Hawkeye predicts are not worth keeping. A variant that always keeps instruction lines removes most of the loss. For Mockingjay, we built a reference model that reproduces the storage of the original design and gives the expected results of our microbenchmarks. Its gem5 implementation is behind schedule, so all Mockingjay results in this report come from the model, and we give a revised timeline.

Index Terms—Cache replacement, last-level cache, Belady's algorithm, gem5

1 Introduction & Problem Statement

The last-level cache (LLC) is the last on-chip structure before DRAM. An LLC miss needs a DRAM access that takes hundreds of cycles, and an out-of-order processor can hide only part of this latency. When the LLC is full, the replacement policy chooses which line to evict, and this choice largely decides the miss rate. The common least-recently-used (LRU) policy fails when a program cycles through a working set larger than the cache, and when data that is never reused pushes useful lines out. Improved policies handle some of these patterns, for example by inserting new lines with a low priority (RRIP [1]) or by learning which load instructions bring in lines that are reused (SHiP [2]). However, each is built around a particular class of access patterns.

Belady's algorithm [3], also called OPT, evicts the line whose next use lies furthest in the future. It is optimal but needs future knowledge. Hawkeye [4] computes what OPT would have done on past accesses and learns which load instructions bring in lines that OPT would keep. Its follow-up, Mockingjay [5], predicts how far in the future each line will be reused, which lets it order lines the way OPT does. Both were evaluated in trace-driven simulators, and gem5 includes neither of them. This project implements both policies in gem5 [6], [7], verifies their functional correctness, and compares their miss rate and performance with each other and with the baseline policies in gem5.

2 Related Work

LRU and LFU rank lines by recency or frequency [8]. RRIP [1] predicts how soon each line will be re-referenced with a 2- or 3-bit counter per line, and DRRIP chooses between two RRIP variants at run time through set dueling [9]. A previous CS 550 project evaluated these policies in gem5, and we use them as baselines. SHiP [2] learns for each program counter (PC) whether the lines it inserts are reused, and SDBP [10] predicts dead blocks from a small sampler. Both learn from a heuristic policy, so they inherit its limits. Hawkeye [4] and Mockingjay [5] both aim to approximate OPT, but they predict different things. Hawkeye learns from OPT's decisions on past accesses, while Mockingjay predicts reuse distances and evicts the line it expects to be reused last. Hawkeye won the 2nd Cache Replacement Championship (CRC2) [11]. We use its CRC2 code and a public ChampSim implementation of Mockingjay as references.

3 Research Questions

- **What are the Hawkeye and Mockingjay policies, and can they be implemented correctly in gem5?**
- **How do their LLC miss rate and IPC compare with each other and with LRU, DRRIP, and SHiP?**
- **Why can they reduce misses where LRU and RRIP fail, and when does predicting reuse distances help more than a keep-or-evict decision?**

4 Hypothesis & Proposed Solution

We hypothesize that learning from OPT lets both policies reduce LLC misses on access patterns where LRU and RRIP fail, such as a working set slightly larger than the cache, or reused data mixed with data that is used only once. We also expect Mockingjay to do better than Hawkeye when lines are reused at different but predictable distances, because it can tell which of the lines worth keeping will be reused last. We do not expect either policy to beat every baseline on every benchmark. Their IPC gains also depend on how much of the miss latency the O3CPU already overlaps. Fig. 1 shows where both policies sit in our gem5 system and their structures.

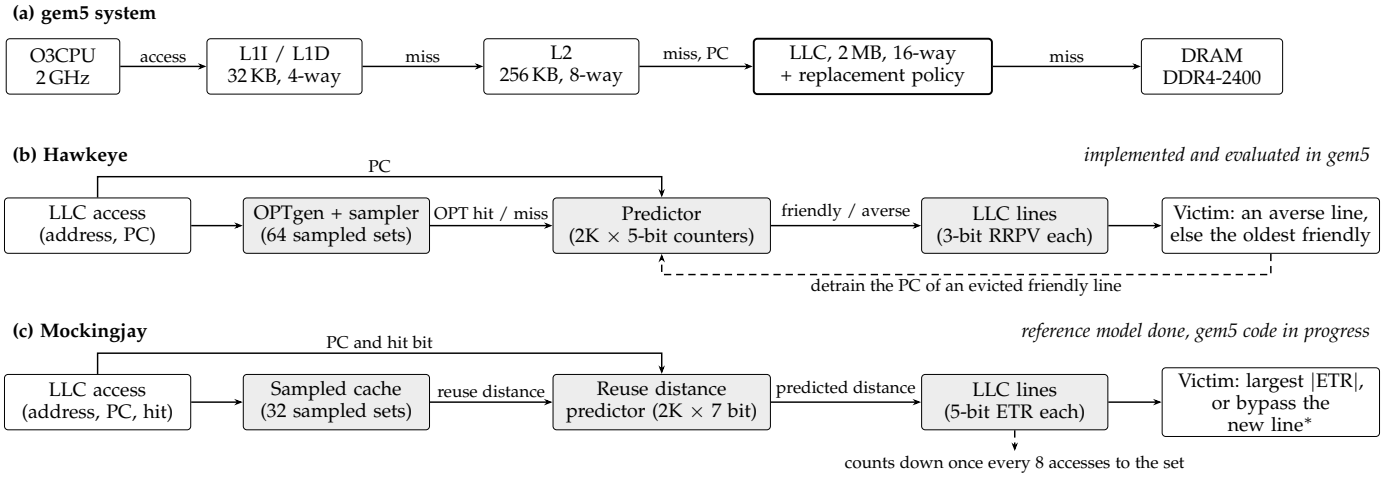


Fig. 1. (a) The gem5 system. Demand requests carry their PC to the LLC. Writebacks and the L2's clean-eviction notices carry none, and the policies handle them separately. (b) Hawkeye and (c) Mockingjay. Shaded boxes are the structures that each policy adds, with sizes for a 2 MB LLC. *gem5 cannot bypass, so our gem5 version inserts the line as the next victim (Section 5.1).

4.1 Hawkeye

OPTgen reconstructs OPT's decisions on a sample of 64 sets. When a line is accessed again, it checks whether OPT could have kept the line since its previous access without exceeding the cache capacity. A **sampler** remembers the PC that last accessed each line of these sets. The **predictor** keeps a 5-bit counter for each hashed PC. It counts up when OPT would have hit and down when OPT would have missed. A line whose PC counter is in the upper half is *cache-friendly* and is inserted with a 3-bit re-reference prediction value (RRPV) of 0. Otherwise it is *cache-averse* and inserted with RRPV 7. Hawkeye evicts an averse line first. If there is none, it evicts the oldest friendly line and, if the line is in a sampled set, lowers the counter of the PC that brought it in.

4.2 Mockingjay

The **sampled cache** records recent accesses to 32 sampled sets. When a line is accessed again, it measures the reuse distance, i.e., the number of accesses to the set since the previous access. A line that is not reused within 127 accesses is treated as never reused. The **reuse distance predictor (RDP)** learns the typical reuse distance of the lines brought in by each PC. Each LLC line keeps a 5-bit **estimated time remaining (ETR)**. It starts at the predicted reuse distance divided by 8 and counts down once every 8 accesses to the set. Mockingjay evicts the line with the largest |ETR|, which is either reused furthest in the future or most overdue. If the incoming line would be reused later than every line in the set, Mockingjay does not insert it.

5 Evaluation

5.1 Methodology

We use gem5 v23.0.0.1 in syscall-emulation mode with the x86 out-of-order CPU model (O3CPU) and the cache hierarchy of Fig. 1(a), without hardware prefetching. For SPEC CPU2017 [12], an atomic CPU skips the first one billion instructions, and the O3CPU then runs 50 million instructions of warmup and 200 million measured instructions. We use the speed versions of mcf, omnetpp, and xalancbmk, and the rate versions of gcc, xz, and deepsjeng, because the speed versions of these three need 7 to 15 GB of memory and the course instance has 15 GB.

Progress made. We added the LLC and the fast-forward step to the course configuration. Hawkeye is a new class (630 lines of C++) in gem5's replacement-policy interface, with a debug flag and statistics. While validating it, we found and fixed three gem5 problems. First, the L2 notifies the LLC of every clean eviction, and gem5 passes these notifications to the replacement policy as if they were accesses. Hawkeye trained on 8.7 million of them in mcf, so it now ignores them. Second, gem5's x86-64 `newfstatat` system call used the wrong file-status structure, so omnetpp and gcc failed to start. Third, gem5 dropped the PC when the L1 forwarded a software prefetch, so 28% of omnetpp's LLC accesses had no PC. We now copy the PC to the forwarded request. After these fixes, the only LLC requests without a PC are the L2's writebacks and clean-eviction notices. As in the CRC2 code, Hawkeye does not train on writebacks. A line filled by a writeback is inserted as cache-averse.

For Mockingjay, we mapped the original design onto gem5's interface. gem5's policies cannot refuse an insertion. However, a line that Mockingjay would not insert gets an ETR larger than every other line in the set, so inserting it normally makes it the next victim. This is an approximation, because the new line still evicts one line. Also, gem5 passes only the accessed line to the policy, so the policy cannot age the whole set every 8 accesses. Each line instead records the set's aging count when its ETR is set, and the current ETR is computed when the line is compared.

5.2 Functional Validation

Progress made. We validate each policy with logs and with microbenchmarks. The Hawkeye debug flag prints every OPTgen decision, prediction, RRPV, and victim. A Python reference model replays the logged LLC accesses through its own OPTgen and compares every decision. It also prints

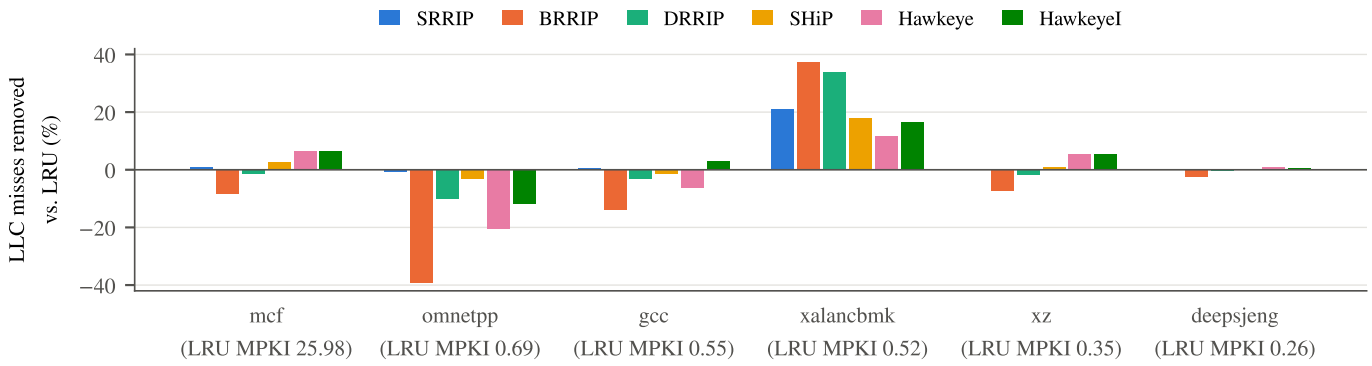


Fig. 2. LLC misses removed relative to LRU with a 2 MB LLC. A negative value means more misses than LRU. Benchmarks are sorted by the LRU miss rate. HawkeyeI always keeps instruction lines.

TABLE 1
LLC miss rate (%) on the microbenchmarks

	Policy	fit	thrash	mix
gem5, 6 passes	LRU, SRRIP	15.0	100	100
	BRRIP	15.0	46.8	96.4
	DRRIP	15.0	61.7	99.1
	SHiP	15.0	90.0	100
	Hawkeye	15.0	48.1	88.3
Model, 6 passes	OPT	14.3	42.9	87.3
	Mockingjay	14.3	54.2	96.4
	no bypass	14.3	55.4	96.4
Model, 20 passes	OPT	4.8	36.5	88.4
	Mockingjay	4.8	41.3	91.2
	no bypass	4.8	41.7	91.2

a short example that can be checked by hand. On a 345,074-access log, gem5 and the model made the same 9,218 OPTgen decisions with no mismatch. This check covers OPTgen only. We have not yet compared the predictor counters, RRPVs, and victims with the model. The log already records them, so we will extend the model to replay them and report any mismatch.

Table 1 shows our three microbenchmarks. *fit* loops over a 1 MB array, *thrash* over a 3 MB array, and *mix* alternates a 1 MB reused array with a stream that is never reused. All policies behave as expected. In fit, every policy has only compulsory misses. In thrash, LRU misses on every access, while Hawkeye keeps part of the array, as BRRIP does. In mix, Hawkeye has the lowest miss rate of the six gem5 policies (88.3%). BRRIP also improves on LRU, but only to 96.4%. For each microbenchmark, the program output is identical under all six policies.

For Mockingjay, we wrote a second reference model that follows the public ChampSim code. It replays the same LLC accesses as our microbenchmarks. The OPT miss rate of these streams (42.9% on thrash) matches the 43.0% that we computed from the gem5 log, so they model the gem5 runs closely. The Mockingjay rows of Table 1 are therefore expected values for the gem5 implementation, not gem5 measurements. They also show two things for our plan. First, in these microbenchmarks, inserting the line instead of bypassing it raises the miss rate by at most 1.2 percentage points. We will check whether this difference stays small on SPEC CPU2017 once the gem5 code runs. Second, Mockingjay learns slowly on short runs. It decides that a line is not reused only after 127 accesses to its set, so on mix it starts keeping the reused array only in the fourth pass. We will therefore run its microbenchmarks for 20 passes. Finally, the model runs the original aging and our gem5 version side by side, and they made the same decision on all 1.36 million accesses.

5.3 Physical Characteristics

We do not implement Verilog, so we estimate the hardware cost from the storage of the added structures. For a 2 MB LLC, our Hawkeye needs 30.0 KB and the paper reports 28 KB. The difference comes from the parameters of the CRC2 code that we follow. For example, its OPTgen records every access instead of every fourth access. Our Mockingjay model needs 31.91 KB, the same as the paper. Both fit in the 32 KB budget of CRC2.

5.4 Performance Impact

Results so far. This section has no Mockingjay results, since Mockingjay does not yet run in gem5. Fig. 2 compares the policies with a 2 MB LLC. On mcf, which has 38 times the LLC misses per kilo-instruction (MPKI) of any other benchmark, Hawkeye removes 6.3% of the misses and improves IPC by 1.4%. No built-in policy improves IPC by more than 0.1%. On xalancbmk, Hawkeye removes 11.5% of the misses, fewer than every built-in policy, while BRRIP removes 37.4%. On omnetpp and gcc, Hawkeye increases misses by 20.4% and 6.2%.

To explain these losses, we added statistics that separate instruction and data lines. In gem5, the PC of an instruction fetch is the fetch address itself, so every instruction line has its own entry in the predictor. On omnetpp, Hawkeye predicted 14% of the instruction fetches as cache-averse, and its LLC instruction misses rose from 662 under LRU to 11,113. On gcc, Hawkeye has fewer data misses than LRU and loses only because of instruction misses. We tested three changes. Refreshing a line when the L2 evicts its copy changed little. Doing so for every line reduced the loss on omnetpp but hurt the other benchmarks. Always inserting instruction lines as friendly (HawkeyeI) brought instruction misses down to the level of LRU or below on every

TABLE 2
LLC misses removed vs. LRU (%) / IPC speedup vs. LRU for three LLC sizes. "Best" is the built-in policy with the highest IPC. A negative value means more misses than LRU.

	LLC	Best baseline	Hawkeye	HawkeyeI
mcf	1 MB	SHiP +7.8 / 1.012	+6.3 / 1.016	+6.3 / 1.016
	2 MB	SRRIP +1.0 / 1.000	+6.3 / 1.014	+6.3 / 1.014
	4 MB	SRRIP +2.7 / 1.015	+10.8 / 1.076	+10.8 / 1.076
xalanc.	1 MB	BRRIP +52.8 / 1.213	+19.5 / 1.061	+22.6 / 1.075
	2 MB	BRRIP +37.4 / 1.023	+11.5 / 1.006	+16.5 / 1.009
	4 MB	DRRIP +2.2 / 1.000	+1.3 / 1.000	+4.5 / 1.001
omnetpp	1 MB	SHiP −15.5 / 1.000	−12.8 / 0.988	−4.1 / 0.995
	2 MB	SHiP −3.0 / 1.000	−20.4 / 0.986	−11.6 / 0.991
	4 MB	SHiP −0.2 / 0.997	−9.9 / 0.989	−5.6 / 0.991

TABLE 3
Milestone status as of Nov. 11

Milestone (planned week)	Status
1. Setup (Week 1, Oct. 29–Nov. 4)	**Met.**
2. Implementation (Week 2, Nov. 5–11)	**Hawkeye met** in Week 1. **Mockingjay late:** design and reference model done, gem5 code by Nov. 18.
3. Functional validation (Week 3, Nov. 12–18)	**Hawkeye mostly met early:** OPTgen check and microbenchmarks done. Predictor, RRPV, and victim check by Nov. 22. **Mockingjay late:** gem5 runs by Nov. 22.
4. Performance evaluation (Week 4, Nov. 19–25)	**Hawkeye met early** at all three LLC sizes. **Mockingjay late:** by Nov. 26.
5. Final report (Week 5, Nov. 26–Dec. 3)	**Not started**, on schedule.

benchmark. As a result, gcc now gains 3.1%, xalancbmk gains 16.5%, and the loss on omnetpp drops to 11.6%. The remaining loss on omnetpp comes from data lines, and we are still analyzing it.

Table 2 repeats the comparison for 1 MB and 4 MB LLCs. On mcf, Hawkeye has the highest IPC at every size, and its advantage grows with the LLC. With 4 MB it removes 10.8% of the misses and improves IPC by 7.6%. On xalancbmk, the working set is between 1 MB and 4 MB, and BRRIP stays well ahead of Hawkeye at 1 MB and 2 MB. Most of this gap comes from data lines, so HawkeyeI closes only a small part of it. On omnetpp, no policy has fewer misses than LRU, and instruction lines cause 71%, 37%, and 34% of Hawkeye's extra misses at the three sizes. HawkeyeI improves IPC over Hawkeye on xalancbmk and omnetpp at every size.

6 Milestones

Table 3 lists our milestones. Milestone 1 is complete. The LLC, the O3CPU, and the baseline policies run in gem5, and every demand access to the LLC carries a PC. The Hawkeye parts of Milestones 2 to 4 are complete two to three weeks ahead of the plan, except the predictor, RRPV, and victim check. Beyond the plan, we also found and fixed three gem5 problems, explained Hawkeye's losses, and tested three variants of it.

We have not met Milestone 2 for Mockingjay. We added Mockingjay on Nov. 2, after the feedback on our proposal, so its work started a week later than Hawkeye's. We also chose to finish the Hawkeye evaluation first, because its analysis exposed gem5 issues that Mockingjay must also handle, i.e., clean eviction notices, missing PCs, and instruction fetches. So far, we have mapped Mockingjay onto gem5's interface and built and checked its reference model. Mockingjay reuses the LLC configuration, scripts, microbenchmarks, and validation flow of Hawkeye, so we plan to complete its gem5 code by Nov. 18, validate it by Nov. 22, and finish its SPEC runs at all three LLC sizes by Nov. 26. This keeps the final report on schedule.

Next steps. Besides the Mockingjay code, we will (i) compare Hawkeye's predictor counters, RRPVs, and victims with the reference model, (ii) compare the Mockingjay gem5 code with its reference model on the same access streams, (iii) measure how much inserting instead of bypassing costs Mockingjay on SPEC CPU2017, and (iv) find the cause of Hawkeye's remaining data-line loss on omnetpp.

7 Division of Labor

Completed so far.

- **Vito Wang (Hawkeye):** implemented Hawkeye in gem5, found and fixed the clean-eviction and software-prefetch problems, wrote the OPTgen reference model and validated Hawkeye against it, analyzed its SPEC CPU2017 results, and tested the three variants including HawkeyeI.
- **Endong Dai (system and baselines):** added the O3CPU, the LLC, and the fast-forward step to the gem5 configuration, fixed the `newfstatat` problem, built the SPEC CPU2017 benchmarks, wrote the microbenchmarks, ran the baseline policies at the three LLC sizes, and collected and plotted the statistics.
- **Xiaotian Tong (Mockingjay model and storage):** wrote the Mockingjay reference model, estimated the storage of both policies, computed the expected microbenchmark results, and mapped the design onto gem5's interface.

Remaining. Vito and Xiaotian will implement Mockingjay in gem5 and validate it against the reference model. Endong will run it with the baselines on SPEC CPU2017 at the three LLC sizes. All three will analyze the results and write the final report.

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